

Communication Lower Bounds for Multi-Party Graph Traversal

Quarter 2 Project Report

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Abstract

A large number of computational problems can be modelled as communication games: For example, given a graph, two players each have a subgraph they know is traversable, and wish to collaboratively construct a path from one vertex to another. Communication complexity studies such situations by creating asymptotic bounds on the minimum communication necessary to complete these tasks. In this report, we prove communication lower and upper bounds for a variety of graph traversal problems, such as the example above. We also review potential applications of these bounds, such as for modelling real-world scenarios and designing algorithms.

Website: <https://rybplayer.github.io/capstone/>
Repository: <https://github.com/rybplayer/DSCCapstone>

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1 Introduction

1.1 Motivating Communication Complexity

As data growth transforms the computational landscape, it will become increasingly difficult practically and computationally for individual machines to compute functions. Practically, proprietary company information may create unexpected and consequential gaps in our data. Say we are trying to use air transit to travel from San Diego to Shanghai, however there does not exist a single airline that has both flights out of San Diego and into Shanghai. Using only one airline, we would be unable to find a route. Do we then conclude that no route exists? No, we understand that by taking different airlines for separate legs of our route we can eventually make it to Shanghai. However, do we have to search through every individual flight or is there a technique that we can employ to reduce the amount of flights that we have to search through?

This is the essence of communication complexity. By studying communication complexity, we are trying to find how much information we need to send from each party in order to compute a function.

1.2 Communication Complexity Basics

Communication complexity studies the communication requirements to solve problems where inputs are distributed among at least 2 parties with unbounded computing power. The base case of a 2 party problem is typically stated as the following:

There are two players, Alice and Bob who each have a n -bit string, a and b that may or may not be the same. The goal is to compute a function, $f(a, b)$ that depends on both a and b .

Under the two party problem, it is always possible to compute $f(a, b)$ by sending all of a to b , effectively reducing problem to a single party. This approach of sending the entire input from one party to another is what we consider the *trivial solution* and is *maximally hard*. We consider this *maximally hard* because it has a communication complexity of $\theta(n)$ bits as all inputs must be sent from a to b . One of the most common examples of a *maximally hard* problem is 2 Player Equality:

Claim The deterministic communication complexity of computing the equality between two parties is $\Omega(n)$.

Proof We can represent the universe of two parties with n bit strings using a $2^n \times 2^n$ boolean matrix M . Given two n bit strings, a and b , $M_{i,j} = 1$ if $a_i = b_j$. When proving equality, if $a = b$ then for all $i = j$, $a_i = b_i$. Therefore, the equality function can also be represented as the identity matrix. It is a

well studied fact that the log of the rank of a boolean matrix representing a function is equivalent to the communication complexity of the protocol. Therefore, since the equality matrix is the $2^n \times 2^n$ identity matrix, whose rank is 2^n , the communication complexity of equality is $\Omega(\log(2^n)) = \Omega(n)$.

		a			
		00	01	10	11
b		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$			
00					
01					
10					
11					

Figure 1: Communication matrix for the equality function on two-bit inputs. The entry is 1 when $a = b$ and 0 otherwise.

By studying Communication Complexity, we seek to find creative ways to achieve non-trivial bounds on multiparty problems or embed proven bounds in novel problems.

1.3 The Graph Traversal Problem

In this report, we study the problem of traversing a graph which is jointly owned by many players. Let there be some graph $G = (V, E)$, where V is the set of all nodes and E is the set of edges. If two or more players “own” some subgraph on the larger graph G , how might they identify if they can use their graphs to collectively construct a path between two arbitrary nodes? Here, we assume that G is both unweighted and undirected.

More formally, let us denote this problem as:

$TRAV(v, w) = \{0, 1\}^n \times \dots \times \{0, 1\}^n \rightarrow \{0, 1\}$ where $TRAV(v, w) = 1$ if there exists a connected path between v and w such that every edge traversed belongs to a graph in $G_1, G_2, \dots, G_k$.

Each subgraph is a connected subset of the entire graph space G , $G_1, G_2, \dots, G_k \subset G$. We consider an edge to belong to a graph if both the starting and ending vertex of the edge are inside of the subgraph.

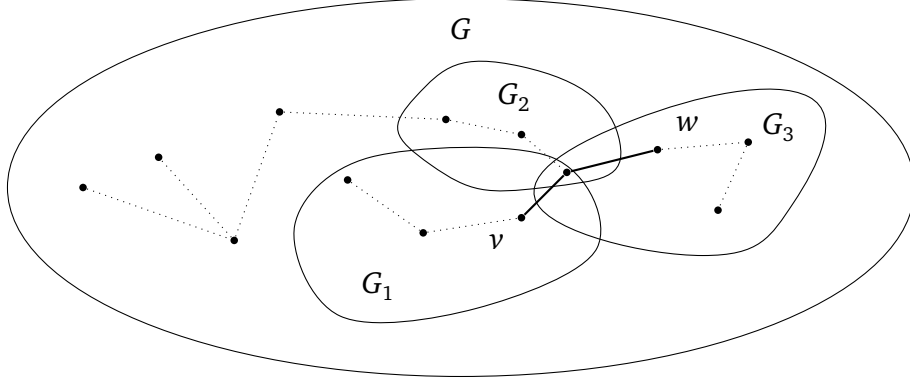


Figure 2: Example graph where $\text{TRAV}(V, W) = 1$

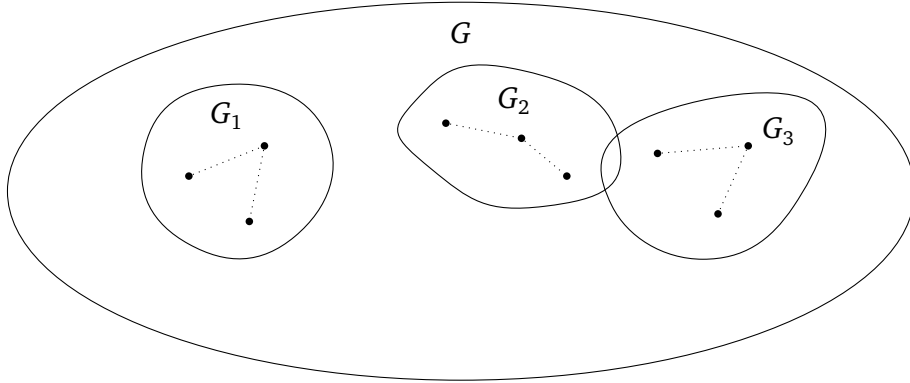


Figure 3: Example where $\text{TRAV}(V, W) = 0$

2 The Two-Party Traversal Problem

2.1 Two-party lower bounds

We begin by considering the case with only two players and thus two subgraphs ($k = 2$). Let these players be named Alice and Bob, with subgraphs G_A and G_B respectively. Denote this problem by TRAV_2 , i.e. the traversal problem for 2 players, where $|V(G)| = n$ and $|E(G)| = O(n^2)$.

Claim 1. Recall that equality EQ has inputs $x, y \in \{0, 1\}^n$, and $\text{EQ}(x, y) = 1$ iff $x = y$. Then $D(\text{TRAV}_2) \geq D(\text{EQ})$.

Proof. **Equality gadget.** We construct a gadget that enforces equality at a single bit position. The gadget is shown in Figure 2.1, and consists of two parallel branches:

1. The top branch corresponds to the assignment $x_i = y_i = 1$,

2. The bottom branch corresponds to the assignment $x_i = y_i = 0$.

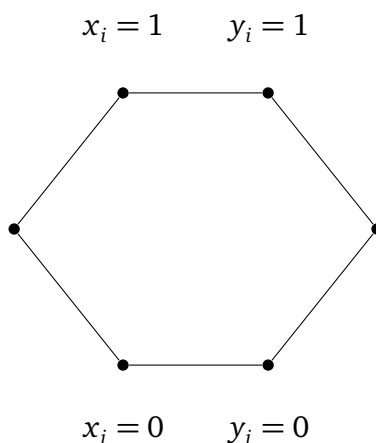


Figure 4: Gadget for proving equality. A path only exists if $x_i = y_i$ for either branch.

Alice enables exactly one entry node of the gadget depending on x_i , and Bob enables exactly one exit node depending on y_i . A path through the gadget exists if and only if Alice and Bob select nodes on the same branch, which holds exactly when $x_i = y_i$.

Reduction to our graph problem. Construct a graph of n equality gadgets in a chain, as in Figure 5. Construct subgraph G_A as follows: Include all vertices at the start and end of each equality gadget. Also, for every $i \in [n]$, give the vertex labelled $x_i = 1$ if this is true, else $x_i = 0$. Then, add all edges between these vertices, as well as all edges between $x_i = z$ and $y_i = z$ for $z \in \{0, 1\}$. Repeat for G_B .

In the resulting graph, there exists a path from v to w if and only if all gadgets are traversable, which occurs if and only if $x_i = y_i$ for all i , and therefore if and only if $x = y$.

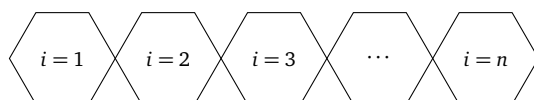


Figure 5: n equality gadgets chained together.

We conclude by noting that the resulting graph G has $\Theta(n)$ vertices and $\Theta(n)$ edges. □

2.1.1 Randomness

Above, we note that the problem is *Deterministically maximally hard*. However, allowing for some error may allow us to produce a random algorithm which performs much better. It can be shown that using the Public Coin Randomness model, Equality—which is Deterministically maximal—can be reduced to being $O(1)$. Unfortunately, for our problem we find that such a bound is unattainable. DISJ, the set disjointness function, has a random communication under both models of $\Theta(n)$, and can be embedded into TRAV.

Claim 2. Recall that in two-party DISJ, Alice and Bob are given sets $X, Y \subseteq [n]$ each. The goal is to determine whether $X \cap Y = \emptyset$. We claim $R(\text{TRAV}_2) \geq R(\text{DISJ})$.

Proof. **Disjointness gadget.** We construct a gadget that enforces disjointness at a single entry $i \in [n]$. The gadget is shown in Figure 6. There, let x_i and y_i be indicator functions whether $i \in X$ and $i \in Y$ respectively. Then:

1. The top branch corresponds to when x_i has the element but y_i does not
2. The middle branch corresponds to when x_i and y_i do not have the element
3. The bottom branch corresponds to when y_i has the element but x_i does not

In other words, there is no path left to right if $i \in X \cap Y$.

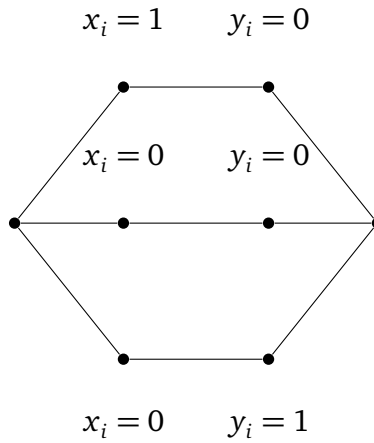


Figure 6: Disjointness gadget. A path left to right only exists if either $x_i = 0$ or $y_i = 0$.

Reduction to our graph problem. We construct a graph G by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex v (left) and end vertex w (right). Construct G_A by giving all vertices at the start and end of every disjointness gadget, and vertices labelled $x_i = 1$ if it is true, else $x_i = 0$. Add all edges between these vertices, and all edges between $x_i = z$ and $y_i = z'$ for $z, z' \in \{0, 1\}$. Do similar for G_B .

In the resulting graph, there exists a path from v to w if and only if all disjointness gadgets are traversable, which occurs if and only if $X \cap Y = \emptyset$.

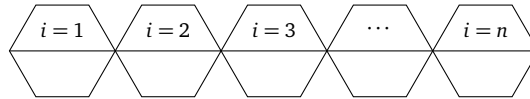


Figure 7: Equality gadgets chained as adjacent hexagonal modules.

We conclude by noting that $|V(G)| = \Theta(n)$ and $|E(G)| = \Theta(n)$. □

2.2 Specialized two-party lower bounds

The previous proofs show that TRAV_2 is maximally complicated in terms of deterministic communication complexity and randomized communication complexity. A natural follow up question is to ask if we can do better. That is, given some additional constraints, can we solve the problem with less communication asymptotically?

Definition 1. Define the CTRAV_2 problem as a TRAV_2 problem where G_A and G_B are guaranteed to be complete and connected.

Claim 3. $D(\text{CTRAV}_2) \geq D(\text{DISJ})$ and likewise for $R(\text{STRAV})$.

Proof. If the start v or end w is in neither G_A nor G_B or both in one of G_A or G_B , the problem can be resolved in $O(1)$ communication. So suppose $v \in G_A$ and $w \in G_B$.

Since G_A and G_B are complete, then a valid path occurs if and only if $V(G_A) \cap V(G_B) \neq \emptyset$. Then, by a similar argument to Claim 2, this problem is at least as hard as DISJ . Thus $D(\text{CTRAV}_2) \geq D(\text{DISJ})$ and likewise for $R(\text{STRAV})$. $\square$

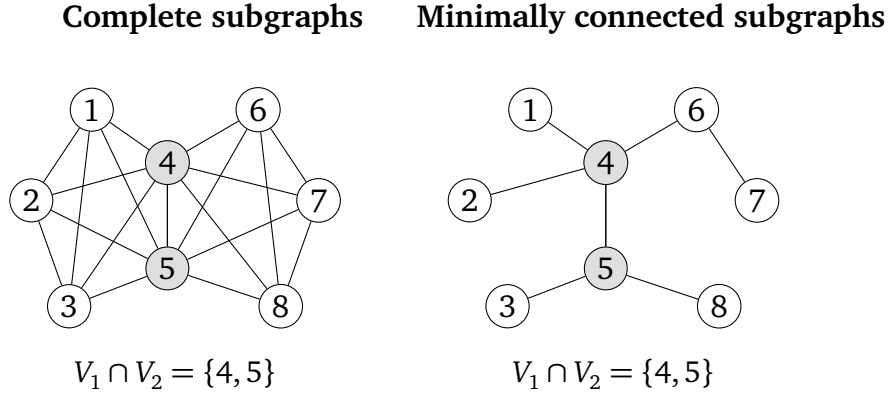


Figure 8: Same vertex sets $V_1 = \{1, 2, 3, 4, 5\}$, $V_2 = \{4, 5, 6, 7, 8\}$ in both panels. Changing edges (complete vs tree) does not change $V_1 \cap V_2$.

This tells us that the local structure of G_A and G_B alone is not a significant source of communication difficulty. That is, even when G_A and G_B are complete graphs, we do not get obviously better lower bounds.

Definition 2. Define STRAV_2 as a TRAV_2 problem where $E(G) = O(n)$, $E(G_A) = O(n)$, and $E(G_B) = O(n)$. In other words, the graph G is sparse and G_A , G_B are sparse too.

Claim 4. $D(\text{STRAV}_2) \geq D(\text{DISJ})$, and likewise for $R(\text{STRAV})$.

Proof. Consider the case where G_A and G_B are connected. If the start v or end w is in neither G_A nor G_B or both in one of G_A or G_B , the problem can be resolved in $O(1)$ communication. So suppose $v \in G_A$ and $w \in G_B$.

Then a valid path occurs if and only if $V(G_A) \cap V(G_B) \neq \emptyset$. Then, by a similar argument to Claim 2, this problem is at least as hard as DISJ. Thus $D(\text{STRAV}_2) \geq D(\text{DISJ})$ and likewise for $R(\text{STRAV})$. $\square$

This tells us that weak edge sparsity of G , G_A , and G_B does not add nor remove significant communication cost.

3 The Multi-Party Traversal Problem

In section 2, we considered TRAV_2 , the communication game where both Alice and Bob computed the minimum length path from v to w using only edges in their subgraphs G_A and G_B . Here, we generalize the problem to TRAV_k , with k players.

The rest of this section is under construction!

Appendices

A Multi-Party Graph Traversal Case Studies

Here, we survey a variety of real-world case studies where communication was a major bottleneck. Note that this section is a work in progress, especially in terms of formal citations.

Air Transportation Large-scale transportation and infrastructure disruptions create a natural model for multi-party graph traversal. Consider the 2010 European volcanic ash shutdown: air-navigation service providers, airports, and airlines each control disjoint portions of airspace and schedules¹. Rerouting passengers is a shortest-path problem in a layered graph whose feasible vertices and edges change over time. No party can broadcast its full real-time state, both for operational and privacy reasons, so route feasibility must be computed with limited information exchange.

Urban Transit Similar graph-traversal constraints appear in urban transit shutdowns and phased restorations, where distinct agencies own stations and tunnels. After Hurricane Sandy, service availability evolved over weeks, and agencies coordinated alternate routes under partial knowledge of what infrastructure was open². Highway detours after the I-95 bridge collapse in Philadelphia illustrate the same phenomenon in a road network: different operators own highway segments and toll roads, and a single failed cut edge forces traffic onto alternative paths with asymmetric knowledge of capacities³.

Maritime Logistics The maritime and logistics examples are equally instructive. The 2021 Suez Canal blockage forced carriers to reroute around the Cape of Good Hope or wait, a decision that depends on schedule and capacity information held privately by shipping lines and port operators⁴. When the Baltimore Key Bridge collapsed, access to the port was blocked and cargo was diverted to alternate ports, again requiring coordination across a sea–port–land graph with private operational constraints⁵.

1. AVIATION.

2. DONOVAN2017333.

3. penndot_i95_2026.

4. 2023OCM...24506868W.

5. pbs_bridge_collapse_2024.

Infrastructure Communication bottlenecks become even sharper in critical infrastructure networks. The Colonial Pipeline cyber disruption similarly forced fuel logistics to reroute through alternate depots and terminals with incomplete visibility⁶. And the 2021 Meta backbone withdrawal⁷ and the 2016 Dyn DNS attack⁸ show that digital services rely on reachability in routing and dependency graphs, where policy and security constraints limit information sharing across administrative domains.

These cases motivate a theoretical study of multi-party graph traversal: agents must collaboratively decide reachability and shortest paths while their private subgraphs, capacities, or weights are not fully shareable. The fundamental question is how much communication is necessary, and how randomized protocols, sketches, or partial disclosures can reduce the burden while preserving correctness.

B Weekly Contributions

We summarize each member’s weekly contributions below:

Table A 1: Weekly contributions by team member.

Week	Ryan Batubara	Ivy Hawks	Darren Ho	Ciro Zhang
6	Work on formally writing up proofs for the trivial and other problems for our final report.	Work on formally writing up the proofs for the two party graph traversal problems.	Work on formally writing up the problem statement and motivation for the problem for our final report.	Work on formally writing up the introduction/motivation for communication complexity for our final report.
5	Help Ivy make the website. Write Github actions to publish to Github Pages. Make subposts components. Wrote CSS.	Start a Github pages website with our latest findings. Convert latex to markdown using Pandoc.	Write up formalized equality reduction in the report. Include the updated simplified procedure from the Professor.	Integrate all notes so far into the report. Fix typos etc in the other people who are writing the report. Give formal definitions, research sections.

6. [crs_colonial_pipeline_2021](#).

7. [cloudflare_facebook_outage_2021](#).

8. [deec66cadd6f410aac1f78a555982100](#).

Week	Ryan Batubara	Ivy Hawks	Darren Ho	Ciro Zhang
4	Introduction section for report, defining communication complexity and various applications of problems.	Look into embedding 3SAT into graph problems. Useful to see techniques to embed hard problems into our problem, especially if we want to prove it is in NP.	Look deeper into random protocols. In particular, develop a bank of random protocols that we can embed into our problem.	Draw figures of our problem and embeddings in LaTeX for the report and poster.
3	Looked at extreme case examples in real-life transport networks, e.g. failure of a node, failure of a “central hub” e.g. in the case on natural disasters, etc.	Wrote down problem statement and preliminary problem solution in latex using hypergraph.	Presented what random protocols are and several different random protocols, namely private and public randomness.	Wrote notes on various variations of shortest path hypergraph problems.
2	Identified various problems that involved the readings that we have found so far. Determined multiparty communication networks with self-driving cars shows promise. Showed how set disjointness can be used to determine collisions with self-driving cars. Also identified a protocol to compute this. Formalized potential problem.	Presented two papers on communication complexity of graph properties to see if there is any information that could be applied to self driving cars. Also presented problems that we will begin working on, with identifying a class of problems where the communication complexity is $O(n)$ by using the blackboard or third party model. Formalized potential problem.	Presented two papers on communication complexity of graph problems. One of them was on determining graph cliques, which can be used to determine whether or not there exists a clique, or a complete network. The other was to show a potential idea on how we can use ideas from other fields of study to compare with communication complexity. Took notes.	Presented a referee model, which we deduced to be the same as the blackboard model. We determined that this idea does not really work for our use case. Identified a critical flaw within our idea that was addressed. Found that our problem was too trivial and by removing/increasing certain assumptions the results remained trivial.

Week	Ryan Batubara	Ivy Hawks	Darren Ho	Ciro Zhang
1	Drafted communication problems with multiparty graph traversal avoiding intersections. Attempted hidden matching problem communication complexity but determined that it would be trivial by having each person explore all nodes.	Drafted communication problems with multiparty weighted graph intersection. Determined that the majority of problems and solutions involved trivial solutions.	Drafted communication problems with regards to black hole problems where there is a node that eliminates all players. Determined that finding it would be trivial by only communicating after 2 or n moves. Recorded meeting notes.	Drafted communication problems where everyone knows about their distance from each other but not any information about the graph. Determined that it would be trivial.

C Updated Proposal

We take a graph, upon which there are players who wish to identify if the graph possesses some property. We wish to use this model to find the minimum communication required for each player to identify if this property exists. Each Player will receive some portion of the graph as their input—this can be either vertices, edges, or both. We mostly study reachability of vertices, in which the only paths considered valid are paths which use only the portions of the graph given to the players as inputs.

For motivation, please see Section 1 of the report.

C.1 Project Goals

Sections 2, ??, 3, and ?? formally define our problem statement and the full spectrum of problems we are considering for our project.

Clearly, section 2 is done. Much of the results for the remaining sections are also complete, and it only remains to formalize and type them up in $\text{\LaTeX}$.

C.2 Actionables

Therefore, this project can be summarized into three specific, actionables:

1. Finish writing this paper to outline the formal, technical communication complexity bounds for a given multi-agent traversal problem with certain constraints.
2. Finish porting the report to our website, whose link is available on the cover page of this report.
3. Give a poster presentation of the key results of the paper, with the intention of making the results applicable to a wider audience.

Therefore, our work will remain theoretically grounded while remaining applicable for the wider computing community.

C.3 Deliverables

We have three main deliverables:

1. A paper presenting actionable item 1 and 2.
2. A research poster of our key results, for actionable 3.
3. A website containing the paper and poster information, for actionable 3.

To be more specific, the paper will take the form of a research paper, where we define the

problems we are working on, prove bounds on these problems, and show how they model situations in the field of multi-agent graph traversal. The paper is our primary project output.

Our detailed weekly progress is shown in the previous appendix section. In the remaining time, we plan for the following actionables for each person:

Table A 2: Weekly plan

Week	Ryan Batubara	Ivy Hawks	Darren Ho	Ciro Zhang
7	Finish the other problems section, with a focus on figures.	Finish the iterated multi-party section, with a focus on figures.	Make significant progress on the multi party section, with a focus on figures.	Make significant progress on the iterated multi-party section, with a focus on figures.
8	Finish the multi-party section.	Finish the iterated multi-party section.	Finish the website port.	Move figures to first draft of poster.
9	Polish paper and proofread website.	Polish website and proofread poster.	Polish poster and proofread paper.	Polish poster and proofread paper.

Indeed, given our weekly contributions thus far we are on track to finishing the project with a week dedicated solely to polishing.